



ReFACT: A Benchmark for Scientific Confabulation Detection with Positional Error Annotations

Yindong Wang* **Martin Preiß*** **Margarita Bugueño**

Jan Vincent Hoffbauer **Abdullatif Ghajar** **Tolga Buz** **Gerard de Melo**

Hasso Plattner Institute / University of Potsdam, Germany

`{yindong.wang,margarita.bugueno,tolga.buz,gerard.demelo}@hpi.de`

`{martin.preiss,janvincent.hoffbauer,abdullatif.ghajar}@student.hpi.uni-potsdam.de`

*Equal contribution

2026 EACL

HUMANE Lab 석사과정생 김태균

26.04.24

Scientific Confabulation

Confabulation is a memory error consisting of the production of **fabricated**, distorted, or misinterpreted memories about oneself or the world.

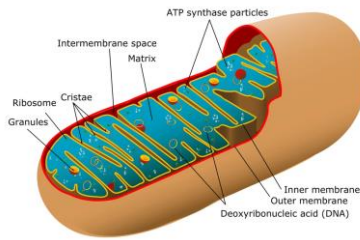


Scientific Confabulation

Confabulation is a memory error consisting of the production of **fabricated**, distorted, or misinterpreted memories about oneself or the world.

Scientific confabulation: generating content that sounds **plausible** to non-experts but is fundamentally factually **incorrect**

ex) "mitochondria produce glucose through cellular respiration"



ReFACT (Reddit False And Correct Text)

A benchmark of 1,001 expert-annotated qa pairs with span-level error annotations

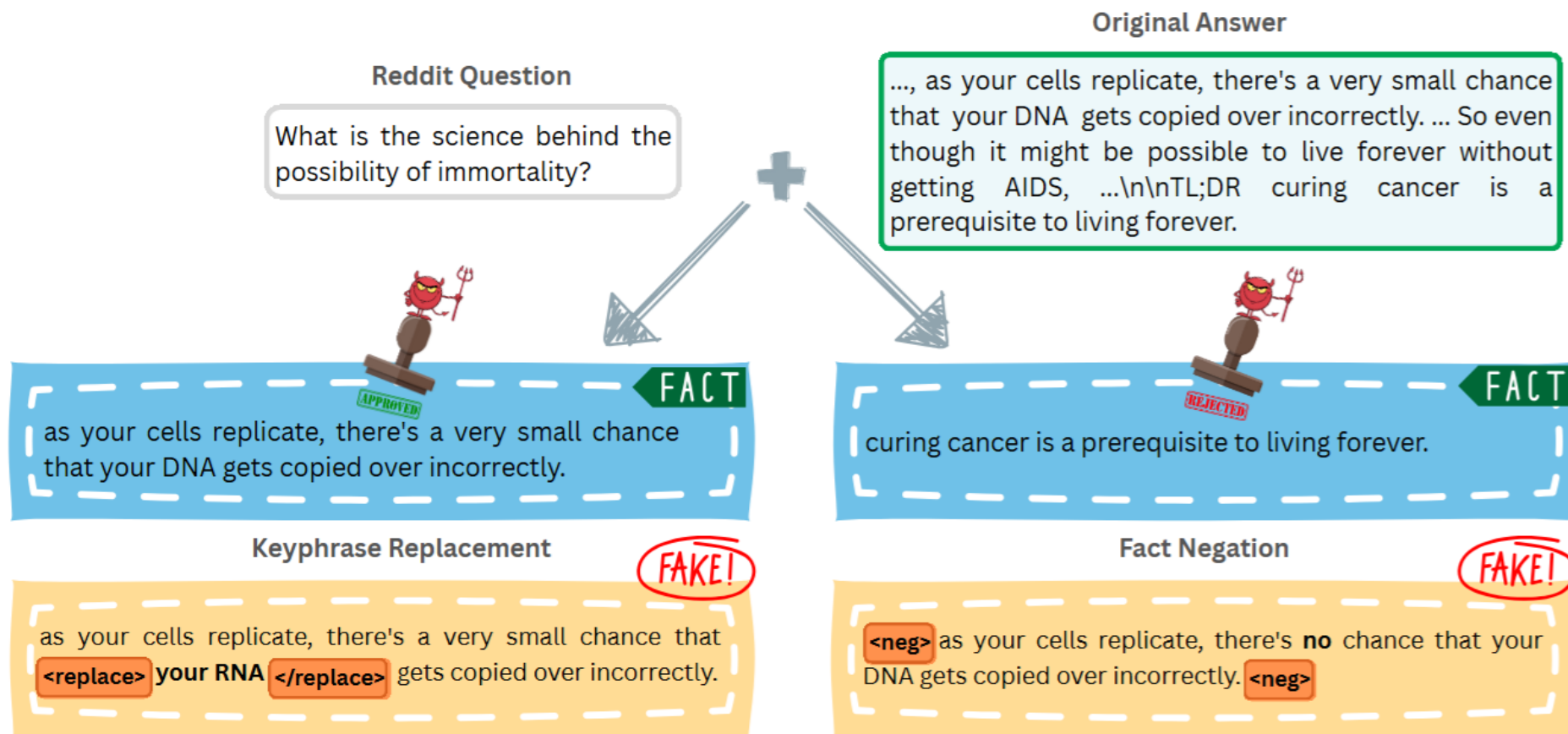
- source: r/AskScience subreddit
- filtering: high-scoring qa pairs, 500-1000 chars

Confabulation Generation

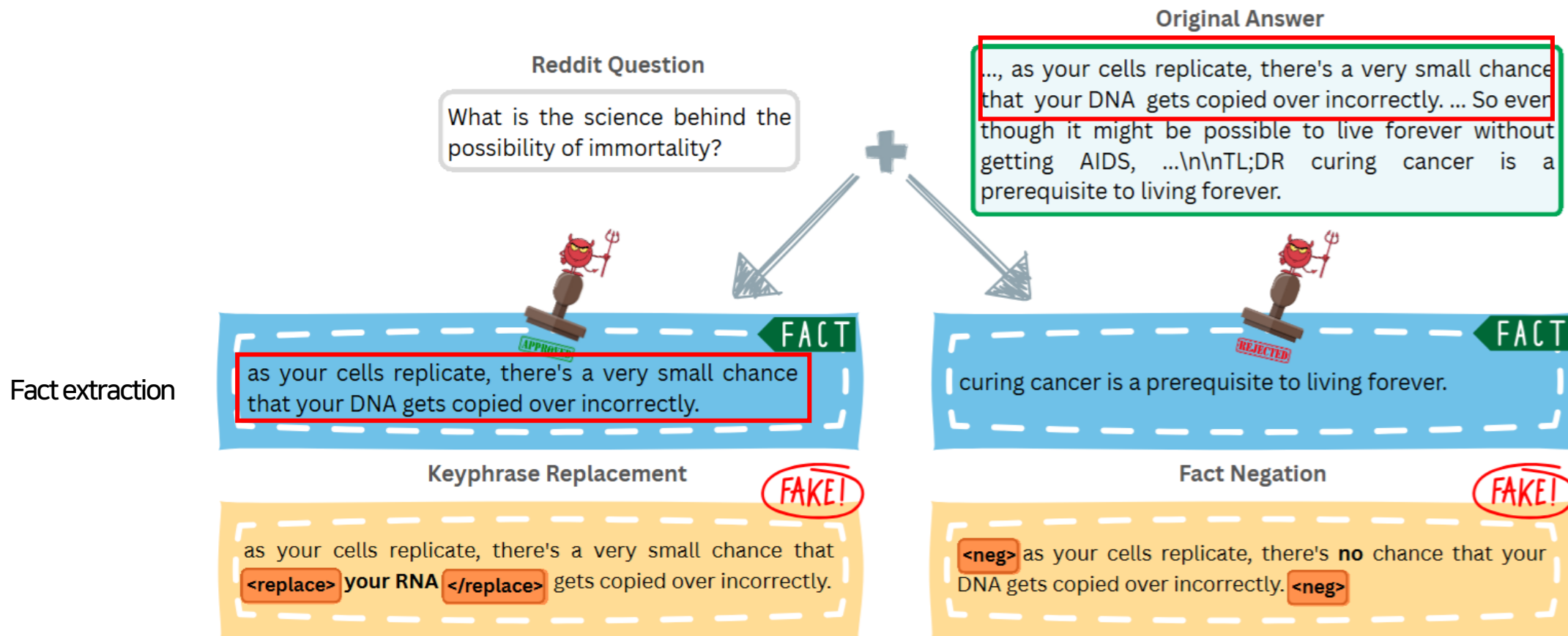
Pipeline:

1. Fact extraction
2. Transformation (**negation** or **entity replacement**)
3. Human verification

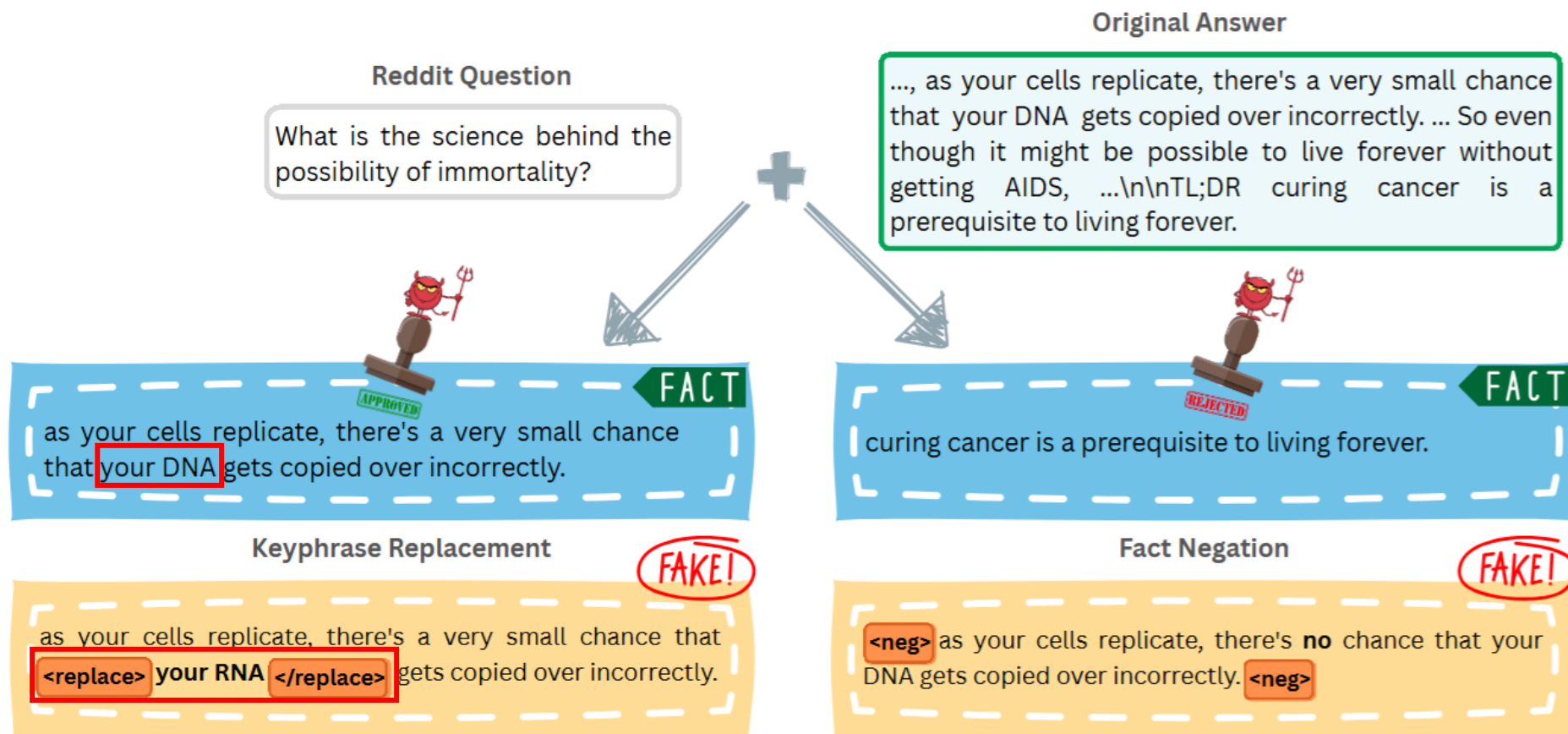
Confabulation Generation



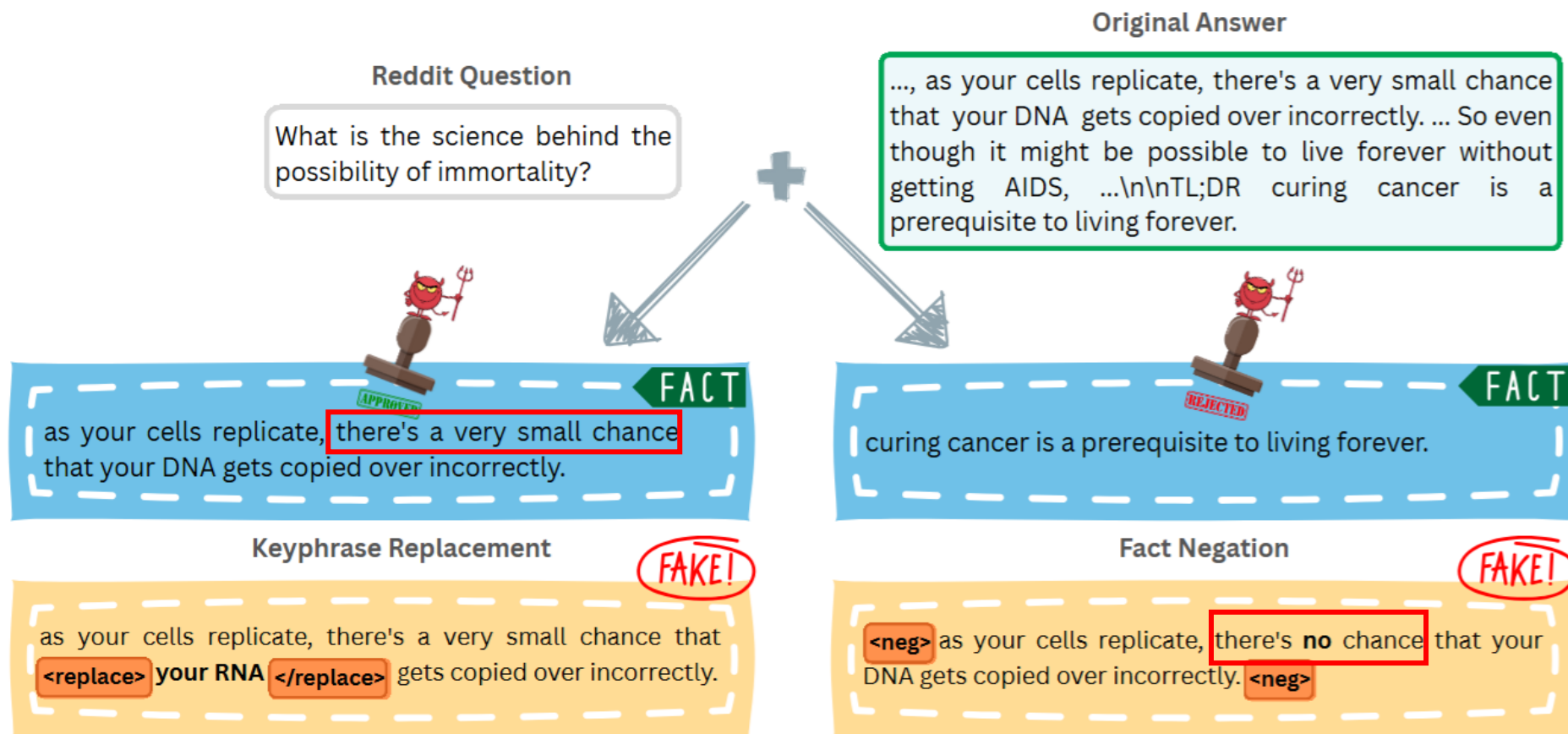
Confabulation Generation



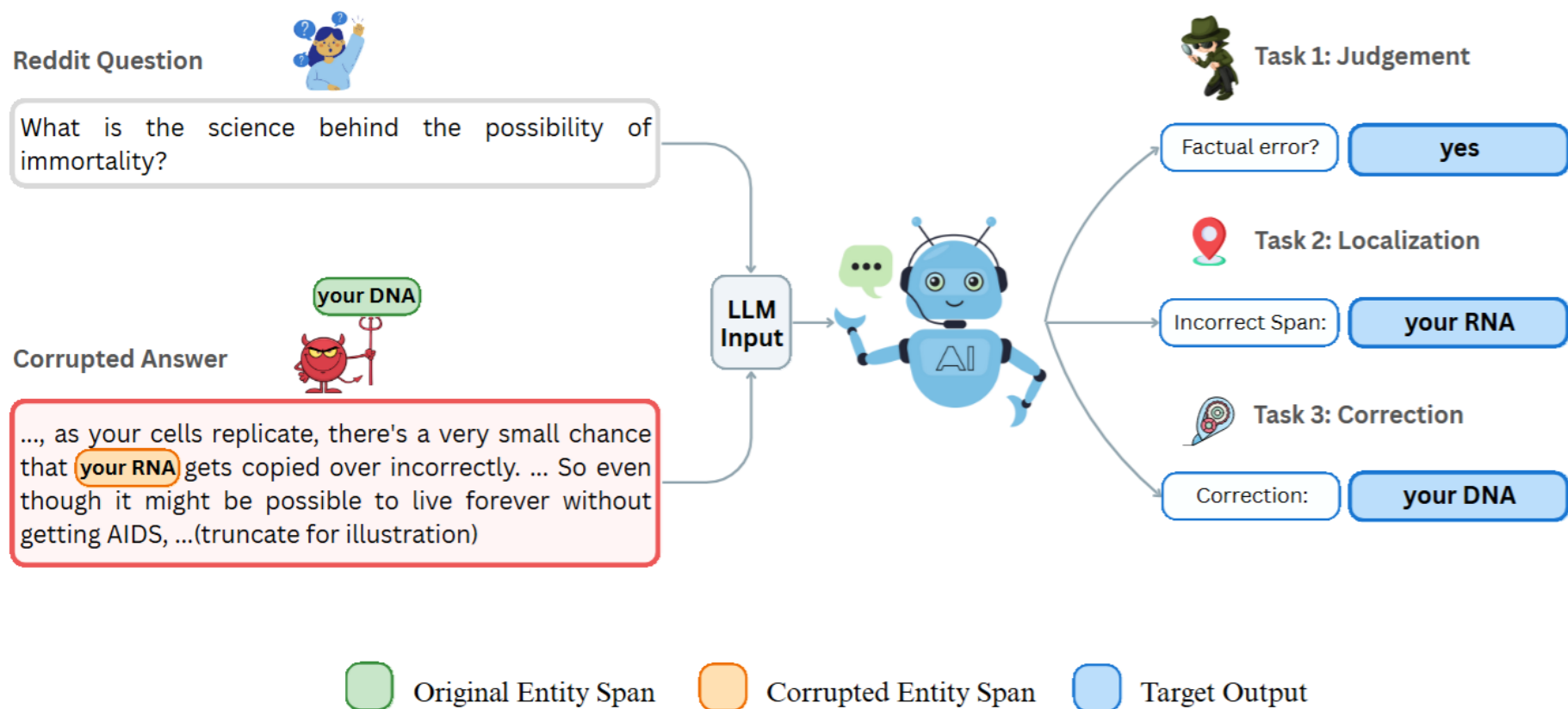
Confabulation Generation



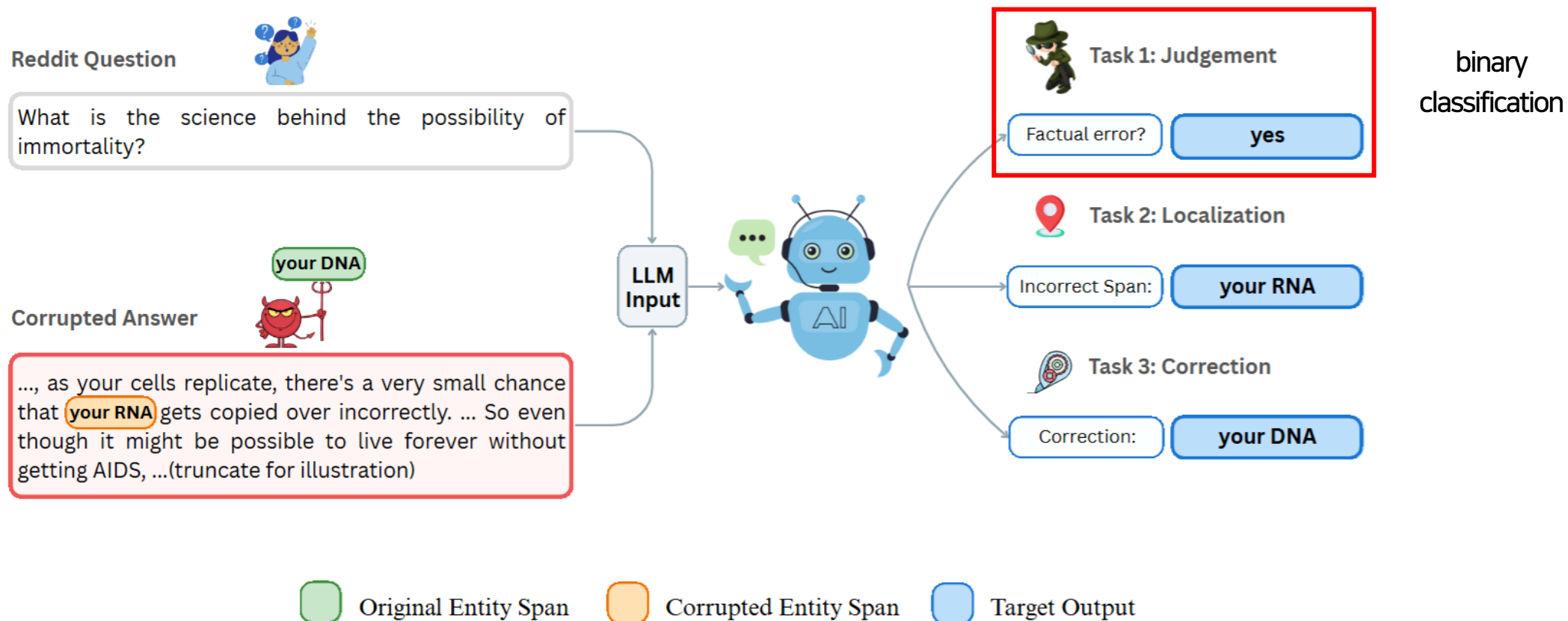
Confabulation Generation



Evaluation Pipeline



Evaluation Pipeline



Judgement Prompt

C.1 Independent Confabulation Judgment

System Prompt:

Your task is to evaluate the factual correctness of a given answer to a question. Read the entire answer and question carefully, then analyze its factual accuracy step by step. After your analysis, provide a final verdict of either “True” if the entire answer is factually correct, or “False” if any part of the answer contains inaccuracies or hallucinations.

User Prompt:

Task:

Question: {question}

Answer: {answer}

Final Verdict:

True / False

C.2 Comparative Confabulation Judgment

System Prompt

Your task is to return the factually correct answer out of the two given answers (A and B) to a question. Read both entire answers and the question carefully, then analyze the factual accuracy of both answers within the context. After your analysis, provide a final verdict of either answer A or answer B is factually correct.

User Prompt

Question: {question}

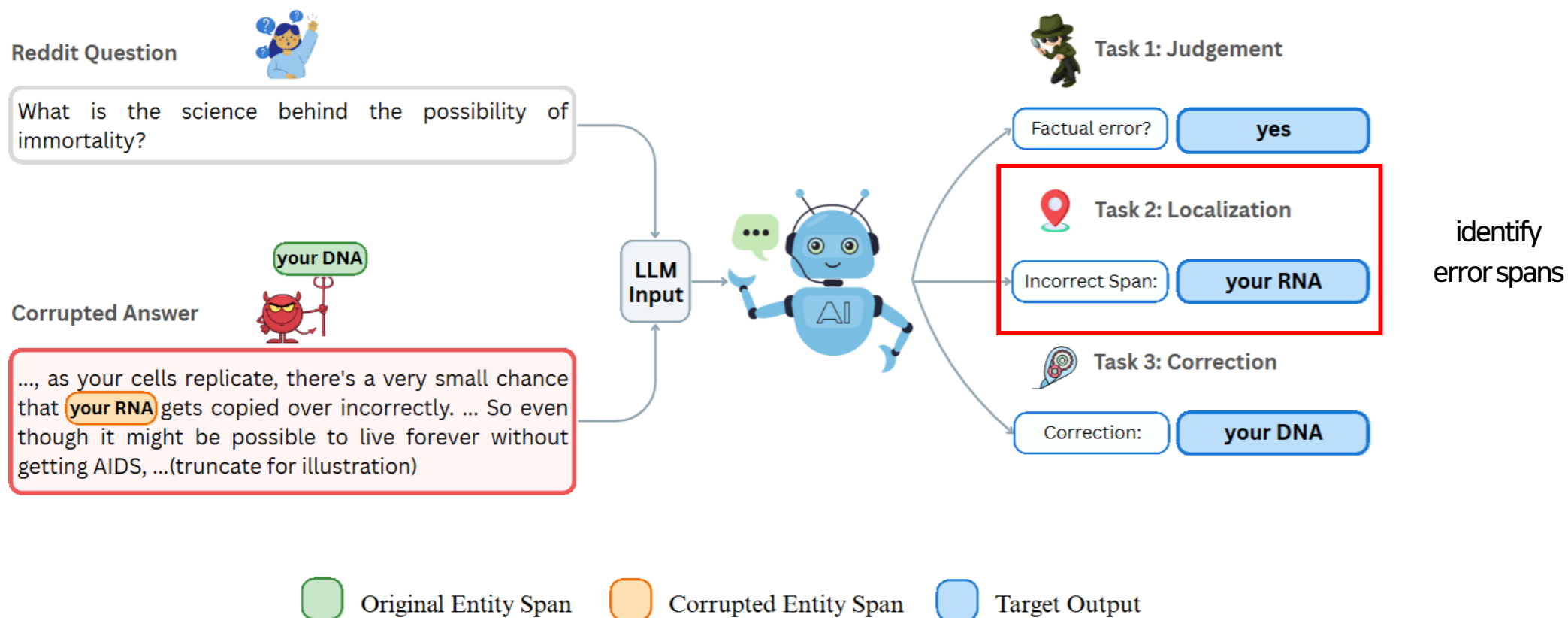
Answer A: {answer_a}

Answer B: {answer_b}

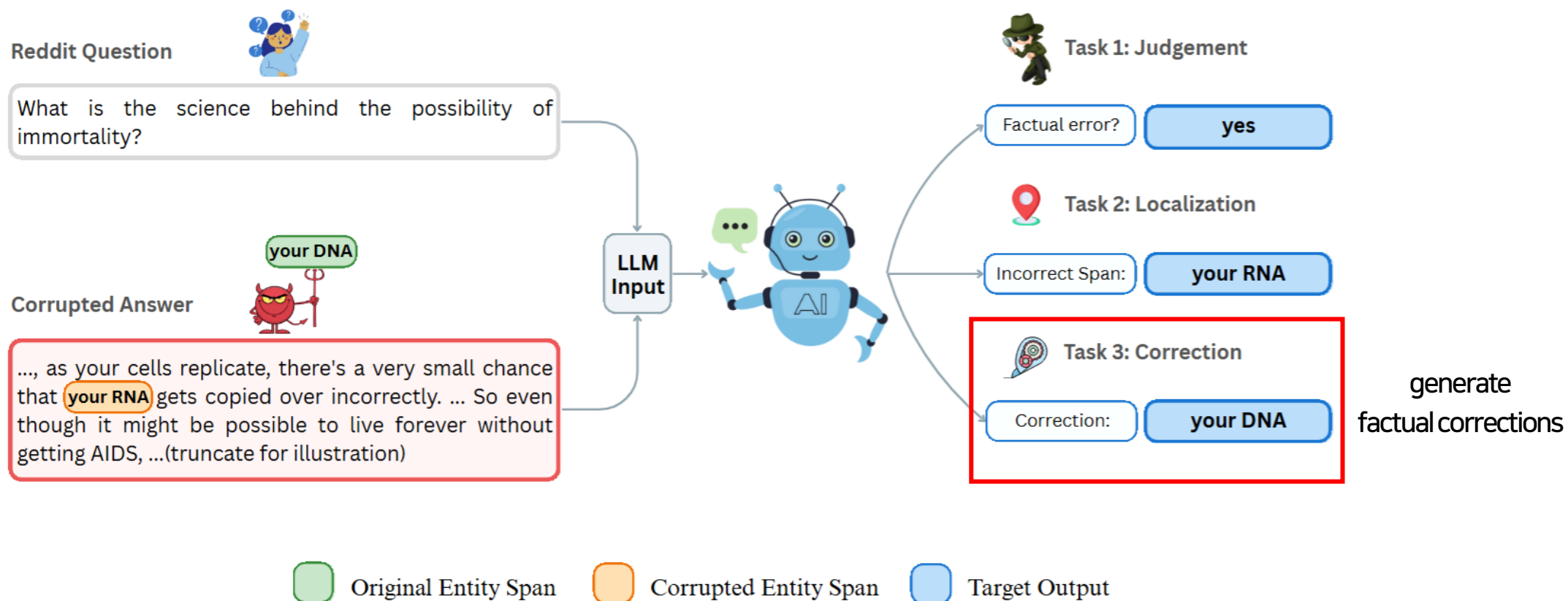
Final Verdict:

answer A / answer B

Evaluation Pipeline



Evaluation Pipeline



Metric

- Aggregate performance
 - Judgement: accuracy, f1
 - Localization: accuracy (IoU)
 - Correctoin: accuracy (exact match)

$$\text{Acc}_{\text{avg}} = \frac{1}{5} \left(\text{Acc}_J^{\text{ind}} + \text{Acc}_J^{\text{comp}} + \text{Acc}_L^{\text{neg}} + \text{Acc}_L^{\text{ent}} + \text{Acc}_C^{\text{ent}} \right)$$

$$\text{Acc}_{\text{loc}} = \frac{1}{N} \sum_{i=1}^N \mathbb{1} \left[\frac{|\hat{s}_i \cap s_i^*|}{|\hat{s}_i \cup s_i^*|} \geq 0.5 \right]$$

$$\text{Acc}_{\text{corr}} = \frac{1}{N} \sum_{i=1}^N \mathbb{1} [\text{norm}(\hat{e}_i) = \text{norm}(e_i^*)]$$

Results

Even sota models struggle significantly with scientific confabulation

Model	Ind. Judg. (Acc / F ₁)	Comp. Judg. (Acc / F ₁)	Neg. Loc. (Acc / IoU)	Ent. Loc. (Acc / IoU)	Ent. Corr. (Acc)	Avg. Acc.
Gemma-3-1B	0.51 / 0.53	0.53 / 0.34	0.13 / 0.13	0.10 / 0.10	0.00	0.25
Gemma-3-4B	0.58 / 0.60	0.52 / 0.33	0.36 / 0.33	0.38 / 0.25	0.10	0.39
Gemma-3-12B	0.65 / 0.63	0.71 / 0.48	0.44 / 0.44	0.46 / 0.26	0.19	0.49
Gemma-3-27B	0.71 / 0.72	0.56 / 0.54	0.61 / 0.46	0.46 / 0.29	0.24	0.52
Llama-3.2-1B	0.49 / 0.39	0.50 / 0.27	0.00 / 0.00	0.08 / 0.04	0.01	0.22
Llama-3.2-3B	0.52 / 0.34	0.48 / 0.29	0.02 / 0.02	0.04 / 0.08	0.02	0.22
Llama-3.3-70B	0.67 / 0.73	0.50 / 0.39	0.69 / 0.61	0.13 / 0.24	0.16	0.43
GPT-4o-mini	0.62 / 0.52	0.59 / 0.55	0.59 / 0.54	0.07 / 0.20	0.21	0.42
GPT-4o	0.67 / 0.67	0.60 / 0.53	0.66 / 0.57	0.47 / 0.38	0.28	0.54

Table 2: Metrics are reported as Accuracy / F₁ or Accuracy / IoU, depending on the task. The final column shows the average of **accuracy** scores only (excluding F₁ and IoU). **Bold** indicates the best-performing model in each metric column.

Results

Data quality or architecture may outweigh sheer parameter scale

Model	Ind. Judg. (Acc / F ₁)	Comp. Judg. (Acc / F ₁)	Neg. Loc. (Acc / IoU)	Ent. Loc. (Acc / IoU)	Ent. Corr. (Acc)	Avg. Acc.
Gemma-3-1B	0.51 / 0.53	0.53 / 0.34	0.13 / 0.13	0.10 / 0.10	0.00	0.25
Gemma-3-4B	0.58 / 0.60	0.52 / 0.33	0.36 / 0.33	0.38 / 0.25	0.10	0.39
Gemma-3-12B	0.65 / 0.63	0.71 / 0.48	0.44 / 0.44	0.46 / 0.26	0.19	0.49
Gemma-3-27B	0.71 / 0.72	0.56 / 0.54	0.61 / 0.46	0.46 / 0.29	0.24	0.52
Llama-3.2-1B	0.49 / 0.39	0.50 / 0.27	0.00 / 0.00	0.08 / 0.04	0.01	0.22
Llama-3.2-3B	0.52 / 0.34	0.48 / 0.29	0.02 / 0.02	0.04 / 0.08	0.02	0.22
Llama-3.3-70B	0.67 / 0.73	0.50 / 0.39	0.69 / 0.61	0.13 / 0.24	0.16	0.43
GPT-4o-mini	0.62 / 0.52	0.59 / 0.55	0.59 / 0.54	0.07 / 0.20	0.21	0.42
GPT-4o	0.67 / 0.67	0.60 / 0.53	0.66 / 0.57	0.47 / 0.38	0.28	0.54

Table 2: Metrics are reported as Accuracy / F₁ or Accuracy / IoU, depending on the task. The final column shows the average of **accuracy** scores only (excluding F₁ and IoU). **Bold** indicates the best-performing model in each metric column.

Results

Smaller models perform near random chance. Larger models show improvement

Model	Ind. Judg. (Acc / F ₁)	Comp. Judg. (Acc / F ₁)	Neg. Loc. (Acc / IoU)	Ent. Loc. (Acc / IoU)	Ent. Corr. (Acc)	Avg. Acc.
Gemma-3-1B	0.51 / 0.53	0.53 / 0.34	0.13 / 0.13	0.10 / 0.10	0.00	0.25
Gemma-3-4B	0.58 / 0.60	0.52 / 0.33	0.36 / 0.33	0.38 / 0.25	0.10	0.39
Gemma-3-12B	0.65 / 0.63	0.71 / 0.48	0.44 / 0.44	0.46 / 0.26	0.19	0.49
Gemma-3-27B	0.71 / 0.72	0.56 / 0.54	0.61 / 0.46	0.46 / 0.29	0.24	0.52
Llama-3.2-1B	0.49 / 0.39	0.50 / 0.27	0.00 / 0.00	0.08 / 0.04	0.01	0.22
Llama-3.2-3B	0.52 / 0.34	0.48 / 0.29	0.02 / 0.02	0.04 / 0.08	0.02	0.22
Llama-3.3-70B	0.67 / 0.73	0.50 / 0.39	0.69 / 0.61	0.13 / 0.24	0.16	0.43
GPT-4o-mini	0.62 / 0.52	0.59 / 0.55	0.59 / 0.54	0.07 / 0.20	0.21	0.42
GPT-4o	0.67 / 0.67	0.60 / 0.53	0.66 / 0.57	0.47 / 0.38	0.28	0.54

Table 2: Metrics are reported as Accuracy / F₁ or Accuracy / IoU, depending on the task. The final column shows the average of **accuracy** scores only (excluding F₁ and IoU). **Bold** indicates the best-performing model in each metric column.

Results

Comparative judgment collapses to near-random performance

Model	Ind. Judg. (Acc / F ₁)	Comp. Judg. (Acc / F ₁)	Neg. Loc. (Acc / IoU)	Ent. Loc. (Acc / IoU)	Ent. Corr. (Acc)	Avg. Acc.
Gemma-3-1B	0.51 / 0.53	0.53 / 0.34	0.13 / 0.13	0.10 / 0.10	0.00	0.25
Gemma-3-4B	0.58 / 0.60	0.52 / 0.33	0.36 / 0.33	0.38 / 0.25	0.10	0.39
Gemma-3-12B	0.65 / 0.63	0.71 / 0.48	0.44 / 0.44	0.46 / 0.26	0.19	0.49
Gemma-3-27B	0.71 / 0.72	0.56 / 0.54	0.61 / 0.46	0.46 / 0.29	0.24	0.52
Llama-3.2-1B	0.49 / 0.39	0.50 / 0.27	0.00 / 0.00	0.08 / 0.04	0.01	0.22
Llama-3.2-3B	0.52 / 0.34	0.48 / 0.29	0.02 / 0.02	0.04 / 0.08	0.02	0.22
Llama-3.3-70B	0.67 / 0.73	0.50 / 0.39	0.69 / 0.61	0.13 / 0.24	0.16	0.43
GPT-4o-mini	0.62 / 0.52	0.59 / 0.55	0.59 / 0.54	0.07 / 0.20	0.21	0.42
GPT-4o	0.67 / 0.67	0.60 / 0.53	0.66 / 0.57	0.47 / 0.38	0.28	0.54

Table 2: Metrics are reported as Accuracy / F₁ or Accuracy / IoU, depending on the task. The final column shows the average of **accuracy** scores only (excluding F₁ and IoU). **Bold** indicates the best-performing model in each metric column.

Results

The correction task remains the most challenging task

Model	Ind. Judg. (Acc / F ₁)	Comp. Judg. (Acc / F ₁)	Neg. Loc. (Acc / IoU)	Ent. Loc. (Acc / IoU)	Ent. Corr. (Acc)	Avg. Acc.
Gemma-3-1B	0.51 / 0.53	0.53 / 0.34	0.13 / 0.13	0.10 / 0.10	0.00	0.25
Gemma-3-4B	0.58 / 0.60	0.52 / 0.33	0.36 / 0.33	0.38 / 0.25	0.10	0.39
Gemma-3-12B	0.65 / 0.63	0.71 / 0.48	0.44 / 0.44	0.46 / 0.26	0.19	0.49
Gemma-3-27B	0.71 / 0.72	0.56 / 0.54	0.61 / 0.46	0.46 / 0.29	0.24	0.52
Llama-3.2-1B	0.49 / 0.39	0.50 / 0.27	0.00 / 0.00	0.08 / 0.04	0.01	0.22
Llama-3.2-3B	0.52 / 0.34	0.48 / 0.29	0.02 / 0.02	0.04 / 0.08	0.02	0.22
Llama-3.3-70B	0.67 / 0.73	0.50 / 0.39	0.69 / 0.61	0.13 / 0.24	0.16	0.43
GPT-4o-mini	0.62 / 0.52	0.59 / 0.55	0.59 / 0.54	0.07 / 0.20	0.21	0.42
GPT-4o	0.67 / 0.67	0.60 / 0.53	0.66 / 0.57	0.47 / 0.38	0.28	0.54

Table 2: Metrics are reported as Accuracy / F₁ or Accuracy / IoU, depending on the task. The final column shows the average of **accuracy** scores only (excluding F₁ and IoU). **Bold** indicates the best-performing model in each metric column.

Results

All models fail to reliably identify specific errors

Model	Ind. Judg. (Acc / F ₁)	Comp. Judg. (Acc / F ₁)	Neg. Loc. (Acc / IoU)	Ent. Loc. (Acc / IoU)	Ent. Corr. (Acc)	Avg. Acc.
Gemma-3-1B	0.51 / 0.53	0.53 / 0.34	0.13 / 0.13	0.10 / 0.10	0.00	0.25
Gemma-3-4B	0.58 / 0.60	0.52 / 0.33	0.36 / 0.33	0.38 / 0.25	0.10	0.39
Gemma-3-12B	0.65 / 0.63	0.71 / 0.48	0.44 / 0.44	0.46 / 0.26	0.19	0.49
Gemma-3-27B	0.71 / 0.72	0.56 / 0.54	0.61 / 0.46	0.46 / 0.29	0.24	0.52
Llama-3.2-1B	0.49 / 0.39	0.50 / 0.27	0.00 / 0.00	0.08 / 0.04	0.01	0.22
Llama-3.2-3B	0.52 / 0.34	0.48 / 0.29	0.02 / 0.02	0.04 / 0.08	0.02	0.22
Llama-3.3-70B	0.67 / 0.73	0.50 / 0.39	0.69 / 0.61	0.13 / 0.24	0.16	0.43
GPT-4o-mini	0.62 / 0.52	0.59 / 0.55	0.59 / 0.54	0.07 / 0.20	0.21	0.42
GPT-4o	0.67 / 0.67	0.60 / 0.53	0.66 / 0.57	0.47 / 0.38	0.28	0.54

Table 2: Metrics are reported as Accuracy / F₁ or Accuracy / IoU, depending on the task. The final column shows the average of **accuracy** scores only (excluding F₁ and IoU). **Bold** indicates the best-performing model in each metric column.

Error Analysis

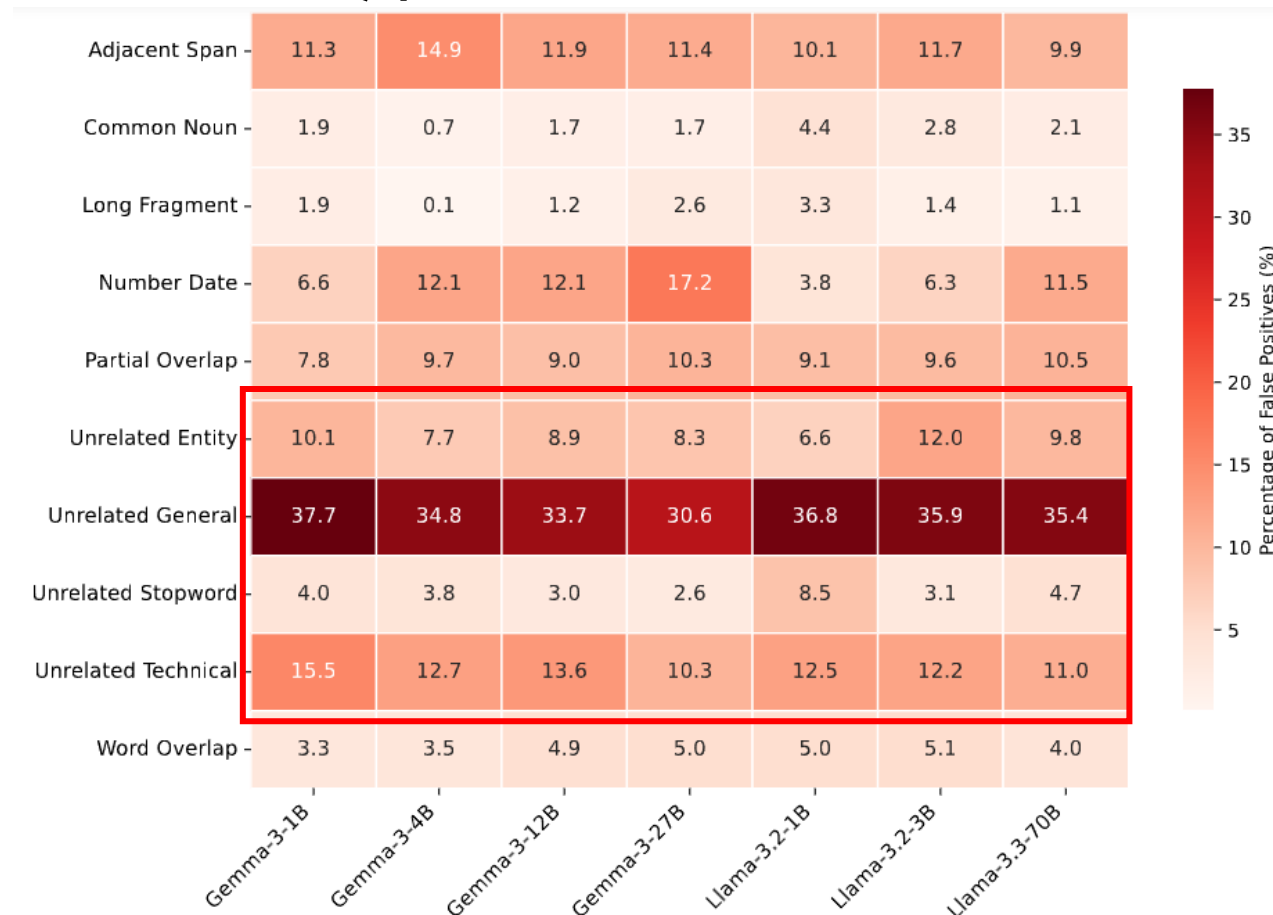
Distractor types on entity localization task

- 1) adjacent_span
- 2) partial_overlap
- 3) word_overlap
- 4) number_date
- 5) common_noun
- 6) long_fragment
- 7) unrelated
 - stopword
 - named entity
 - technical term
 - general content



Error Analysis

Models fixate on **salient distractors** contextually plausible

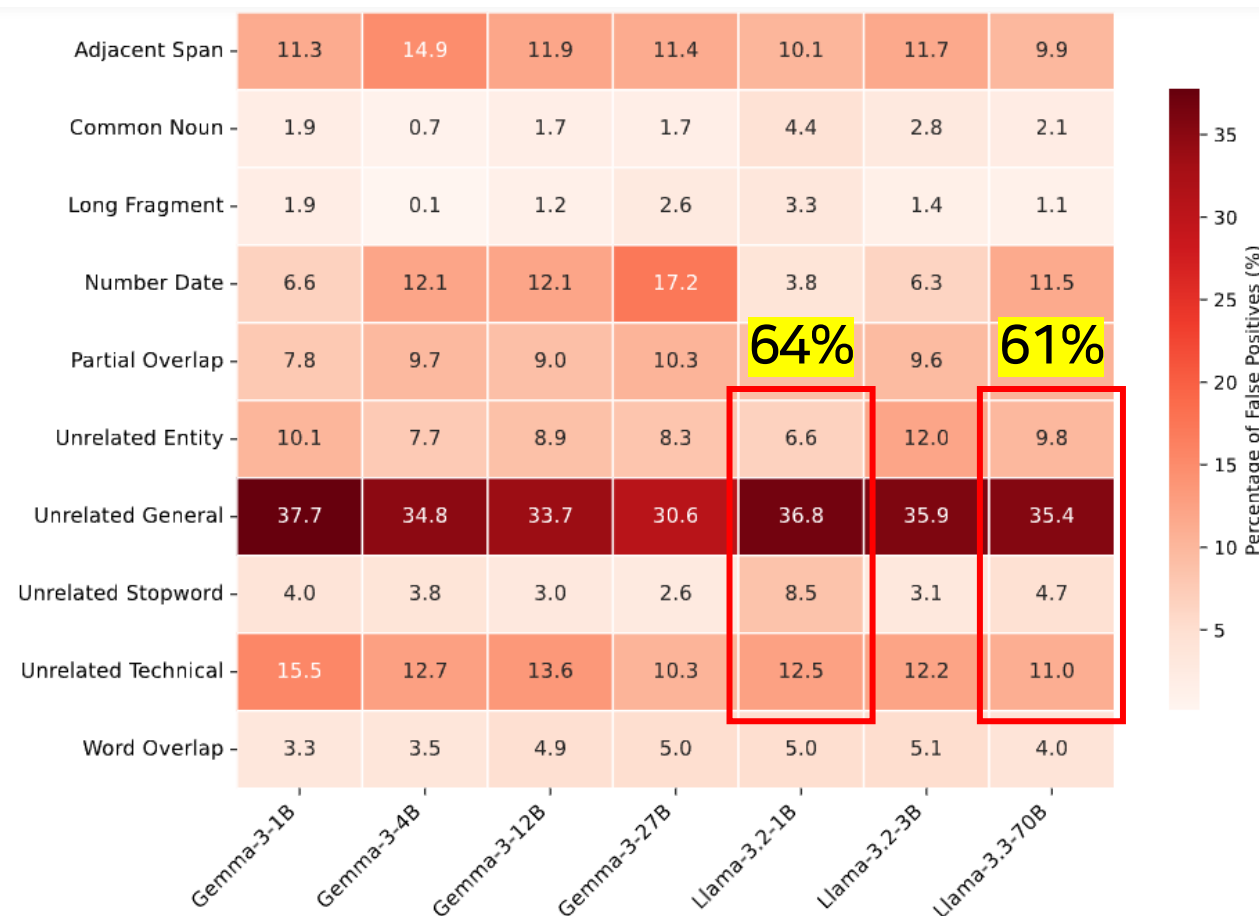


Error Analysis

Invariance to scale and prompting

1) scaling limitation

2) prompting failure



Error Analysis

Invariance to scale and prompting

1) scaling limitation

2) prompting failure

Model	Zero-Shot	Few-Shot	Few-Shot + CoT
Gemma-3-1B	0.504	0.455	0.432
Gemma-3-4B	0.584	0.561	0.517
Gemma-3-12B	0.653	0.629	0.612
Gemma-3-27B	0.707	0.704	0.640
Llama-3.2-1B	0.475	0.416	0.463
Llama-3.2-3B	0.481	0.412	0.487
Llama-3.3-70B	0.651	0.653	0.644

Prompting strategy ablation for independent judgement

Zero-shot consistently matches or outperforms few-shot and CoT prompting
=> bottleneck is not instruction adherence or reasoning depth, but the underlying representation of scientific truth

models cannot "reason" their way out of a knowledge deficit

Key Findings

- Comparative judgment collapses to near-random performance
- Domain-specific substitutions are substantially harder to detect than logical negations
- Correction remains largely unsolved

What makes ReFACT unique?

- Combines real community Q&As with systematic transformations and full human verification.
- Provides span-level error annotations and enables three evaluation tasks (detection, localization, correction) - not just binary classification.

Contributions

- Scientific confabulation benchmark
- Three-task evaluation
- The salient distractor phenomenon

개인적인 생각

- 데이터셋 구축 과정에서 사용한 LLM에 대한 보조 실험이 없음
- 실제 사람(전문가) 평가의 부재
- error analysis 각 유형의 분포 및 예시, 분류 방법 등 구체적인 내용이 부록에도 없음
- 직관과 반대되는 결과가 나오더라도 결과 자체를 해석할 수 있는 능력을 키우자